

Objective

- To study the effect of repeated stimuli and production of fatigue in skeletal muscle using power lab .

Requirement:

- Nerve muscle preparation
- Lab tutor, Stand
- Ringer's solution or normal saline
- pins and dissecting instrument.

□ In this practical we shall learn about

1-Muscle fatigue

2-Summation

3-Tetanzation

1-What is fatigue

- When a muscle is repeatedly stimulated, it becomes gradually less excitable and ultimately ceases to respond.
- This phenomenon is called fatigue.

Types of Fatigue

- Transmission Fatigue or indirect Fatigue
- Contraction Fatigue, Muscular Fatigue or Direct Fatigue

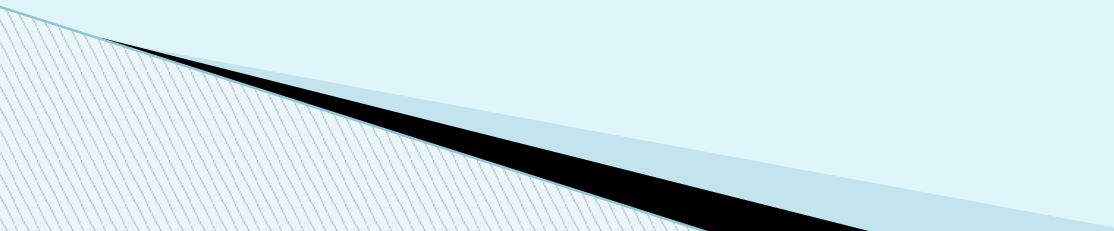
Transmission Fatigue or indirect Fatigue

This type of fatigue is produced when the muscle is indirectly stimulated i.e. by placing the electrodes on the nerve.

When the repeated stimulus is applied to the muscle through the nerve, all the stored acetylcholine which is a chemical transmitter for the transmission of nerve impulse from the nerve to the muscle at the neuromuscular junction is exhausted.

Thus the nerve impulse can not be transmitted to the muscle at the neuromuscular junction.

Hence the site of fatigue is neuromuscular junction and not the muscle itself.

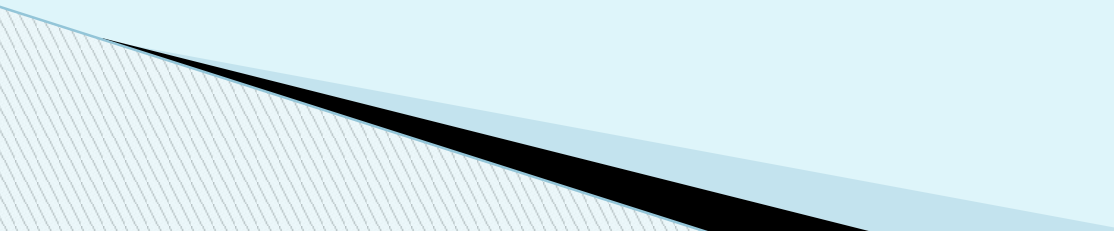


Contraction Fatigue, Muscle Fatigue, or Direct Fatigue

This type of fatigue is recorded by placing the electrodes directly on the muscle and stimulating it repeatedly.

When the muscle contracts repeatedly (exercise) a metabolic end product **lactic acid** accumulated in the contractile tissue fibers causing failure of contraction. This is also due to the depletion of energy.

In frogs, the isolated muscle can not recover from fatigue but in human muscles the formation of lactic acid is a reversible phenomenon and giving rest to the muscle it can recover and lactic acid in presence of oxygen is converted back to pyruvic acid.



Summation

To stimulate the muscle with two pulses at different pulse intervals and observe the effect on muscle contraction.

summation

The additive effect of several electrical impulses on a neuromuscular junction, the junction between a nerve cell and a muscle cell.

Individually the stimuli cannot evoke a response, but collectively they can generate a response.

Theory

When two maximal stimuli (S1) are applied in such a way that second stimuli comes while the muscle is still facing the first then the effect of second stimulus(S2) will add up with the 1st ;
This will increase the height of contraction, this is called the summation effect on SMT

When second stimulus applied in different phases of muscle contraction following response will result.

S2 in latent period (LP) of the 1st SMT, the effect of second stimulus does not show any response, because muscle remains in refractory period during this period

S2 in Contraction period (C>P)

Here the effect of second stimulus summates with the first one and it causes the increase in height of contraction.

S2 in relaxation period (Early)

When second stimulus applied in early relaxation phase, the second response will be produced by second stimulus which has slightly higher peak of contraction.

S2 in Relaxation Period (R.P)

When second stimulus falls in late R.P the two separate curves will be obtained.

Tetanzation

Tetanization, in a physiological context, refers to the process of inducing or the state of having tetanic contractions (sustained muscle contractions) by stimulating a muscle's motor unit with a high frequency of impulses.

When repeated stimuli are applied with high frequencies to a muscle fused twitches are recorded, known as tetanization

When the frequency is such that each successive stimuli falls within the relaxation phase it is called indulation and the graph recorded is called incomplete tetanization. The indulation decreases with the increase in frequency till it disappears and a straight line is recorded due to the sustained contraction of the muscle. This response is called as complete tetanization.

Relevant Questions

What is meant by summation?

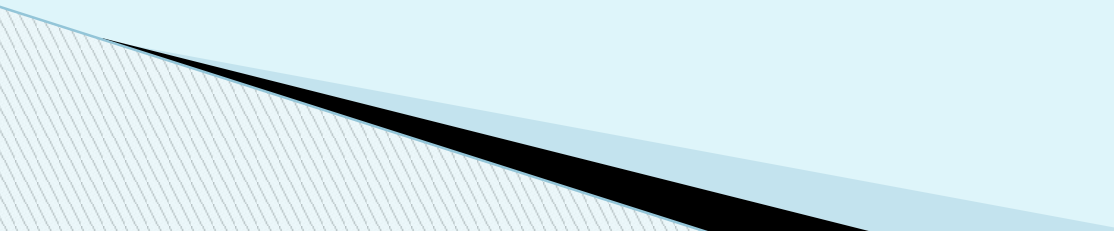
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What is meant by motor unit?

The fundamental functional unit of skeletal muscle, consisting of a single motor neuron and all the muscle fibers it innervates, responsible for generating muscle contraction.

What is motor unit recruitment

Motor unit recruitment is the process by which different motor units are activated to produce a given level and type of muscle contraction.



How can we increase the tension in the muscle? Explain in terms of multiple fiber summation and the frequency summation

To increase muscle tension, you can employ multiple fiber summation (recruiting more motor units) and frequency summation (increasing the firing rate of motor neurons), both of which lead to stronger muscle contractions.

Increasing or decreasing the number of motor units active at any one time changes the amount of force produced by a muscle.

How does fatigue occur in a skeletal muscle

the limitations of a nerve's ability to generate a sustained signal (neural fatigue); and the reduced ability of the muscle fiber to contract (metabolic fatigue).

Reductions in intramuscular levels of ATP, phosphocreatine and glycogen, and low blood glucose (hypoglycemia) can impair skeletal muscle performance.

What is difference between tetanisation, tetanus and tetany?

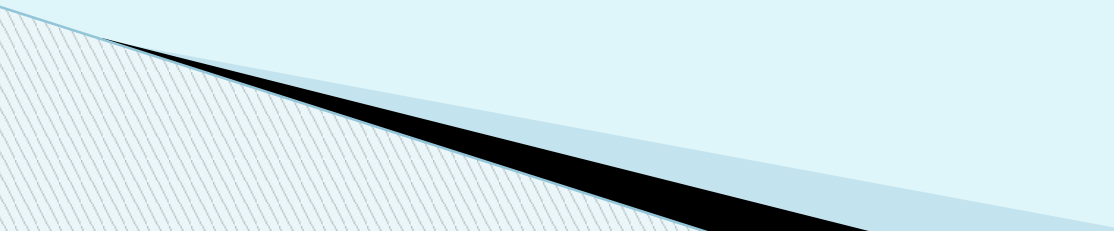
Tetanzation is a physiological process, tetany is a symptom of neuromuscular excitability, and tetanus is a disease caused by a bacterial infection.

Is the contraction time the same for all muscles

No, contraction times vary significantly between different muscles and muscle fiber types, with fast-twitch fibers contracting and relaxing more quickly than slow-twitch fibers.

Difference red and white muscle fibers

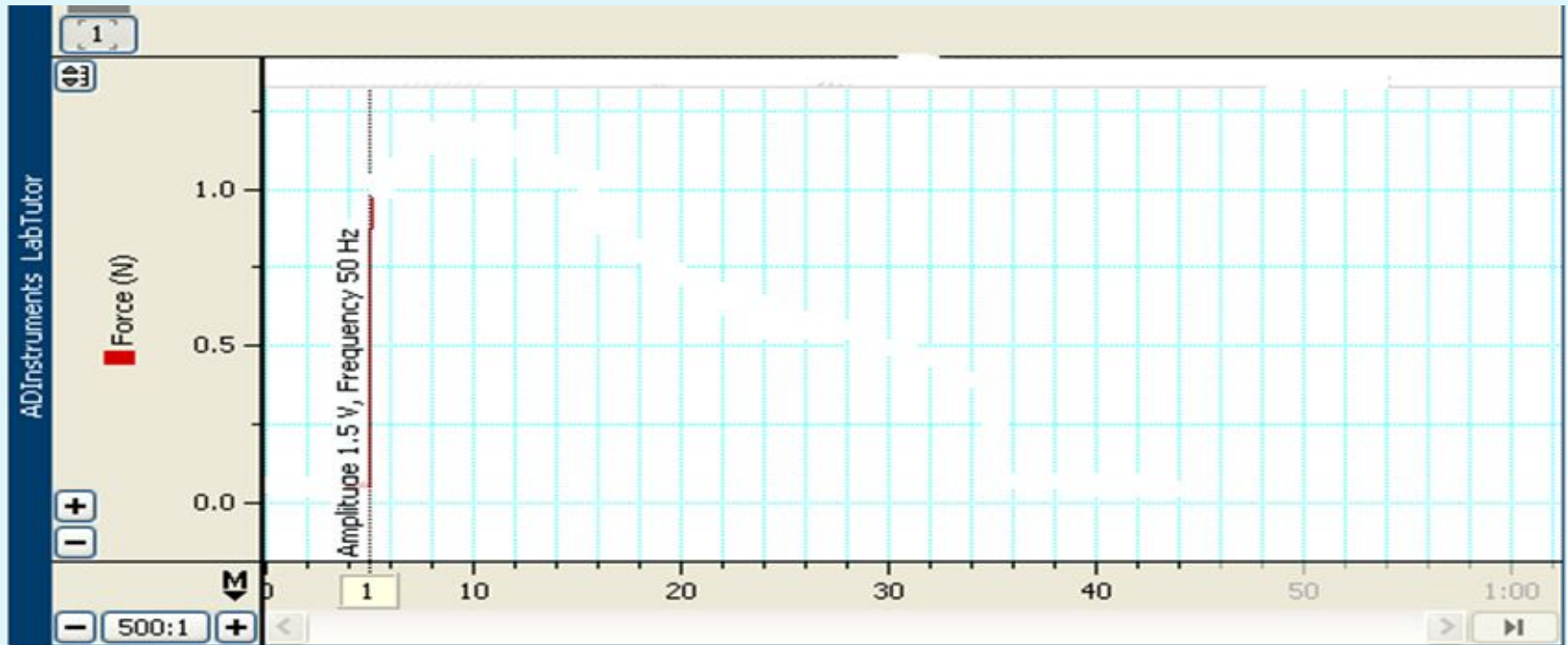
Red muscles are named so because they are dense with capillaries and are rich in myoglobin and mitochondria – which gives it a characteristic red appearance. On the other hand, white muscles have comparatively less mitochondria and are myoglobin poor, giving the muscles a “whitish” appearance.



Distinguish between isometric and isotonic contraction

Isometric contractions are contractions in which there is no change in the length of the muscle. No joint or limb motion occurs. Isotonic contractions occur when the muscle changes length, producing limb motion. Concentric contractions occur when the muscle shortens.

Power lab



Value	Force

Table	% Decline in Force		
	Maximal Force (N)	Force at End of Stimulation (N)	Decline in Force from Maximum (%)

Procedure

- Make sure the muscle is moist and the stimulator handle is positioned correctly.
- in the stimulator panel, insert your value for the supramaximal stimulus voltage the you calculated in Exercise 1.
- click start. After 5 seconds of baseline recording, Lab Tutor will stimulate the muscle for 30 seconds with pulses 20 ms at intervals.
- Recording will continue for a further 10 seconds after the last stimulus.

Analysis

- You will have seven records to analyze.
- For each record, place the marker on the baseline immediately before stimulation begins.
- Use the Waveform Cursor to measure the maximum force of contraction for the record.
- Click to transfer the value into the Value panel.
- Drag the value to the appropriate place in the Table

Thank You